



DeepFake Shield

Multimodal AI-Powered Digital Media Forensics & Deepfake Detection Platform

Detect. Localize. Explain. Investigate.

DeepFake Shield is an AI-powered digital media forensics platform designed to detect, analyze, localize, and explain manipulated or AI-generated **images, videos, and audio**.

Instead of simply predicting whether a piece of media is "real" or "fake", DeepFake Shield combines **deep learning, temporal analysis, explainable AI, metadata analysis, compression analysis, and media provenance** to produce an evidence-based forensic report.



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1. Project Overview

The rapid growth of generative AI has made it increasingly easy to create convincing synthetic media.

Deepfake technology can be used to manipulate:

- Faces
- Voices
- Videos
- Images
- Expressions
- Identity
- Lip movements
- Backgrounds

This creates challenges in:

- Journalism
- Social media
- Cybersecurity
- Digital investigations
- Identity verification
- Fraud detection
- Misinformation prevention

DeepFake Shield addresses this problem by creating a **multimodal media-analysis platform** capable of examining multiple forms of digital evidence.

2. Problem Statement

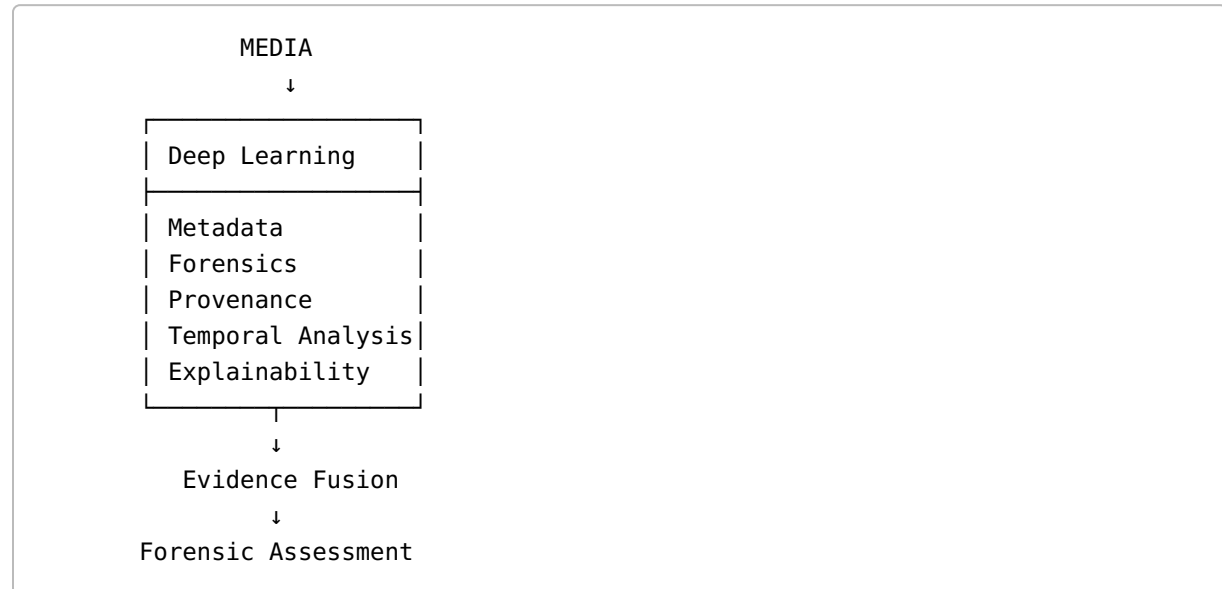
Traditional deepfake detection systems often have a narrow objective:

```
graph TD
    A[Input Media] --> B[ML Model]
    B --> C[REAL / FAKE]
```

This provides little information about:

- Why the media was classified as fake
- Which region was manipulated
- Which frames are suspicious
- Whether the manipulation is consistent across time
- Whether metadata has been modified
- Whether the file contains suspicious compression patterns
- Whether provenance information is available

DeepFake Shield aims to solve this by producing a more comprehensive analysis.



3. Motivation

A detector should not simply say:

"This video is fake."

It should provide evidence such as:

"The media is likely manipulated. The facial region contains anomalous visual patterns, several consecutive frames exhibit temporal inconsistencies, and the available metadata provides no reliable provenance information."

The goal is therefore:

Detection + Localization + Explanation + Evidence

4. Proposed Solution

DeepFake Shield will provide a web-based interface where users can upload:

- Images
- Videos
- Audio files

The platform performs multiple levels of analysis.

```
graph TD;
    Upload --> Media[Media Identification];
    Media --> Preprocessing;
    Preprocessing --> Modality[Modality-Specific Analysis];
    Modality --> Deep[Deep Learning Detection];
    Deep --> Forensic[Forensic Analysis];
    Forensic --> Explainability;
    Explainability --> Evidence[Evidence Fusion];
    Evidence --> Final[Final Assessment];
    Final --> Report[Interactive Report];
```

5. Core Objectives

Primary objectives

1. Detect AI-generated and manipulated media.
 2. Support image, video, and audio analysis.
 3. Localize suspicious regions.
 4. Identify suspicious video frames.
 5. Analyze temporal inconsistencies.
 6. Provide explainable predictions.
 7. Analyze metadata and compression artifacts.
 8. Provide evidence-backed forensic reports.
 9. Provide confidence and risk scores.
 10. Build a deployable web platform.
-

6. Key Features

Detection

Detect whether media is likely:

- Authentic
 - Manipulated
 - AI-generated
 - Inconclusive
-

Localization

Identify:

- Manipulated facial regions
 - Suspicious image areas
 - Suspicious video frames
 - Potentially synthetic audio segments
-

Explainability

Provide:

- Heatmaps
- Attention maps

- Frame-level scores
 - Important features
 - Model confidence
-



Temporal Analysis

Analyze videos frame-by-frame and across time.



Forensic Analysis

Analyze:

- Metadata
 - Compression
 - Image artifacts
 - Face boundaries
 - Lighting consistency
 - Temporal consistency
-



Provenance

Check available media provenance information.

Where supported, the system can inspect provenance/authenticity metadata rather than assuming that missing metadata means a file is fake.



Forensic Report

Generate a complete analysis report containing:

- Final verdict
 - Confidence
 - Evidence
 - Suspicious regions
 - Suspicious frames
 - Metadata
 - Model results
 - Analysis timeline
-

7. Supported Media

Image

Supported formats:

JPG
JPEG
PNG
WEBP

Video

Potential formats:

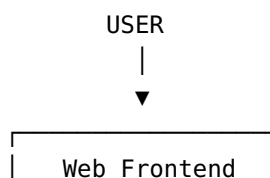
MP4
MOV
AVI
WEBM

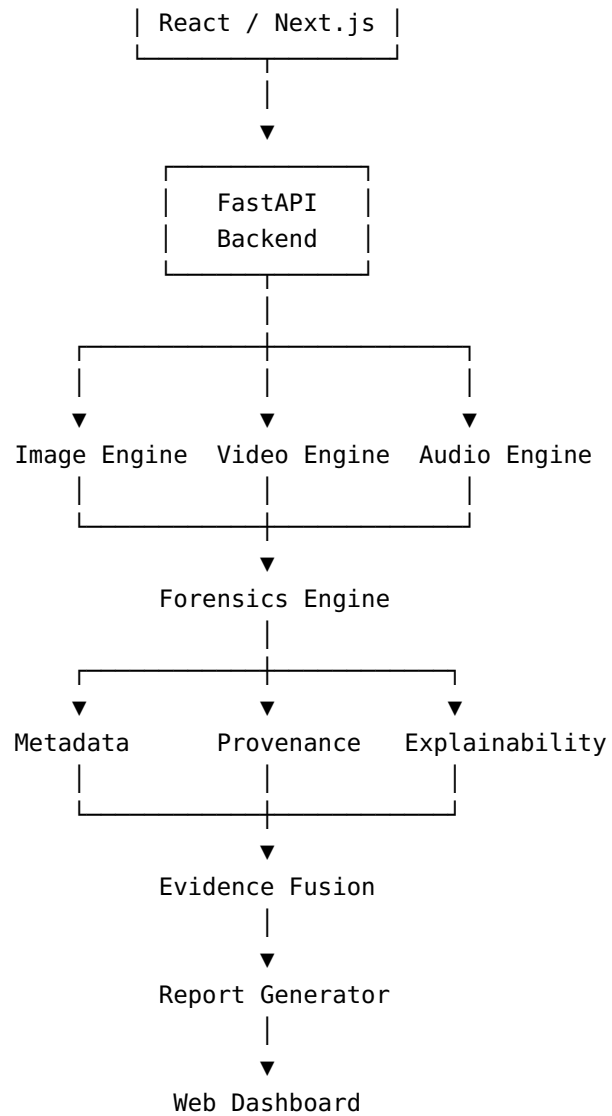
Audio

Potential formats:

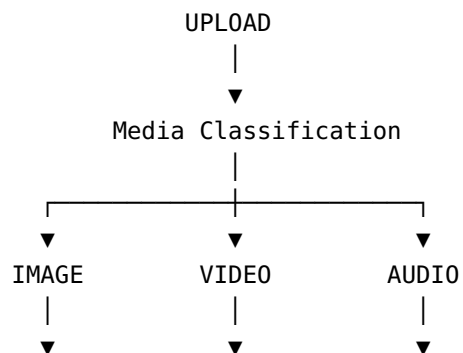
WAV
MP3
FLAC
M4A

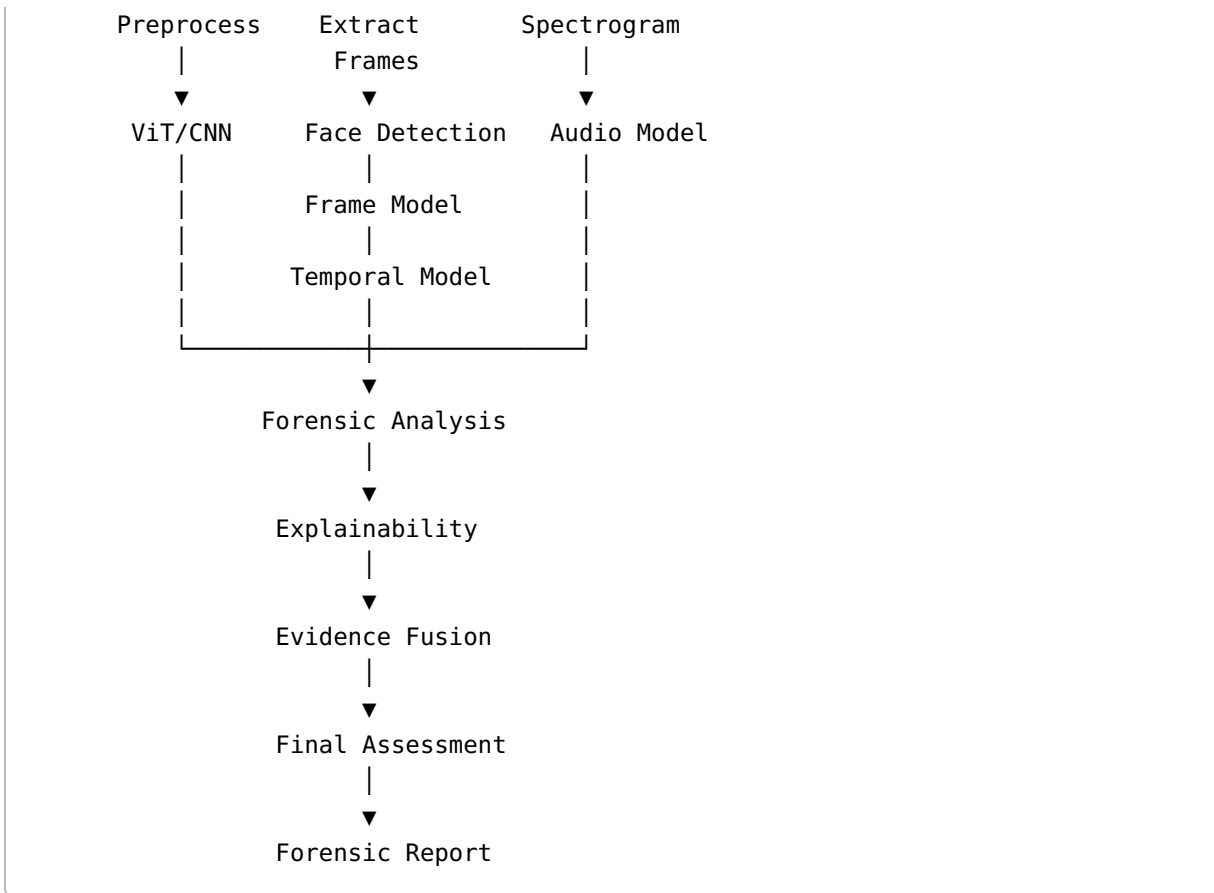
8. System Architecture





9. Complete Processing Pipeline





10. Image Deepfake Detection

The image pipeline:

```
Image
↓
Image Validation
↓
Face Detection
↓
Face Alignment
↓
Face Crop
↓
Resize / Normalize
↓
Deep Learning Model
↓
Prediction
```

↓
Explainability

Model candidates

Start with:

- EfficientNet
- ConvNeXt

Then experiment with:

- Vision Transformer
- Swin Transformer

The project should compare different architectures rather than assuming that one model is automatically best.

Example

```
Input:
portrait.jpg

Prediction:
LIKELY MANIPULATED

Probability:
91.7%

Suspicious region:
Face boundary

Secondary region:
Eyes / Mouth
```

11. Video Deepfake Detection

Video detection is more complex because temporal information matters.

Pipeline:

```
Video
↓
Frame Sampling
↓
Face Detection
↓
Face Tracking
↓
Face Alignment
↓
Frame-Level Model
↓
Temporal Model
↓
Video-Level Prediction
```

Frame-level analysis

Example:

```
Frame 001 → 0.08
Frame 002 → 0.11
Frame 003 → 0.16
Frame 004 → 0.63
Frame 005 → 0.81
Frame 006 → 0.92
Frame 007 → 0.88
```

The system can visualize this as a timeline.

Temporal model

Frame embeddings:

```
F1
F2
F3
F4
F5
F6
```

↓

Temporal Transformer

↓

Video Authenticity

The temporal model attempts to capture inconsistencies that may not be obvious from a single frame.

12. Audio Deepfake Detection

Audio processing:

```
Audio
↓
Noise normalization
↓
Segmentation
↓
Mel Spectrogram
↓
Feature Extraction
↓
CNN / Transformer
↓
Synthetic Speech Prediction
```

Potential features:

- Spectral characteristics
 - Prosody
 - Pitch patterns
 - Energy
 - Temporal characteristics
-

13. Explainable AI

DeepFake Shield should not be a black box.

Image explanation

Generate:

```
Original Image
      +
Prediction
      +
Heatmap
```

Possible techniques:

- Grad-CAM
- Integrated Gradients
- Attention visualization

Video explanation

Display:

```
00:03  _____  ⚠
00:04  _____  ⚠
00:05  _____  ⚠
00:06  _____
```

Users can click a suspicious timestamp and inspect the frame.

14. Digital Forensics Engine

The forensic engine performs analysis independent of the main neural classifier.

Metadata analysis

Extract available metadata such as:

```
File type
File size
Creation timestamp
```

Modification timestamp

Image dimensions

Video codec

Frame rate

Audio codec

Software tags

Important:

Missing metadata does NOT automatically mean manipulation.

Metadata should be treated as evidence, not proof.

Compression analysis

Analyze:

- JPEG artifacts
 - Recompression patterns
 - Blocking artifacts
 - Quantization patterns
 - Inconsistent compression
-

Image artifact analysis

Potential indicators:

- Facial boundary inconsistencies
- Texture anomalies
- Lighting inconsistencies
- Color inconsistencies
- Unnatural reflections
- Blending artifacts

These are **signals**, not definitive proof.

15. Provenance Analysis

DeepFake Shield can inspect available provenance information.

Possible sources include:

- Embedded metadata
- Content credentials
- Cryptographic provenance information
- C2PA-compatible information where available

The system should distinguish between:

Verified provenance
Unknown provenance
Conflicting provenance
Missing provenance

It must **never interpret missing provenance as proof of a deepfake.**

16. Evidence Fusion

This is one of the project's most important components.

Instead of:

Deep Learning → 92% fake

combine multiple signals.

Example:

Deep Learning Score	0.91
Temporal Anomaly	0.84
Forensic Artifact Score	0.72
Metadata Signal	0.20
Provenance Signal	0.10

These can be presented as separate evidence signals.

A fusion model can eventually learn how to combine them.

17. Risk & Confidence Score

Instead of making a legally definitive claim, DeepFake Shield should produce an **assessment**.

Example:

AUTHENTICITY ASSESSMENT

LIKELY MANIPULATED

Model confidence: 91.7%

Evidence:

- Visual anomalies
- Temporal inconsistencies
- Face-region anomalies

Possible categories:

LIKELY AUTHENTIC
LIKELY MANIPULATED
LIKELY SYNTHETIC
INCONCLUSIVE

18. Dataset Strategy

Potential datasets for experimentation include:

FaceForensics++

Useful for manipulated facial videos.

Celeb-DF

Useful for higher-quality deepfake videos.

DFDC

Large-scale deepfake detection dataset.

DeeperForensics

Useful for robustness/generalization experiments.

ForgeryNet

Large-scale forgery dataset with multiple manipulation types.

For audio, dedicated spoof/synthetic-speech datasets should be used.

19. Machine Learning Pipeline

```
Dataset
↓
Data Cleaning
↓
Duplicate Detection
↓
Train / Validation / Test Split
↓
Face Detection
↓
Face Alignment
↓
Augmentation
↓
Model Training
↓
Validation
↓
Hyperparameter Optimization
↓
Testing
↓
Cross-Dataset Testing
```

20. Model Architecture

Version 1 — Image

```
Image
↓
Face Detector
↓
Face Crop
↓
EfficientNet / ConvNeXt
↓
Binary Classifier
↓
Real / Fake
```

Version 2 — Image Transformer

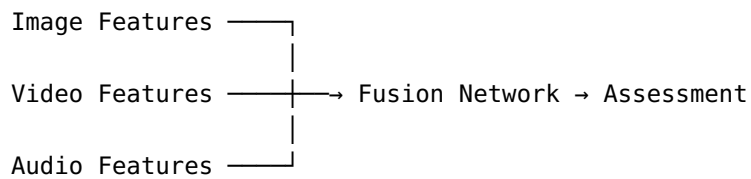
```
Face Crop
↓
Vision Transformer
↓
Classification Head
↓
Real / Fake
```

Version 3 — Video

```
Video
↓
Frames
↓
Face Detection
↓
ViT
↓
Frame Embeddings
↓
Temporal Transformer
```

↓
Video Classification

Version 4 — Multimodal



21. Training Strategy

Start simple.

Phase 1

Train:

EfficientNet

on image/face crops.

Phase 2

Compare:

EfficientNet
ConvNeXt
ViT

Phase 3

Build video-level detection:

ViT
+
Temporal Transformer

Phase 4

Add audio.

Phase 5

Add multimodal fusion.

22. Evaluation

Do NOT rely only on accuracy.

Use:

- Precision
 - Recall
 - F1-score
 - ROC-AUC
 - PR-AUC
 - False Positive Rate
 - False Negative Rate
 - Calibration
-

Important experiment

Cross-dataset generalization

Train:

Dataset A

Test:

Dataset B

This tests whether the model learned actual manipulation patterns rather than dataset-specific artifacts.

23. Web Application

Frontend:

Next.js / React
TypeScript
Tailwind CSS

Main pages

1. Home

DEEPFAKE SHIELD

Analyze digital media using AI-powered
forensic analysis.

[Upload Media]

2. Upload

Drag & Drop Media

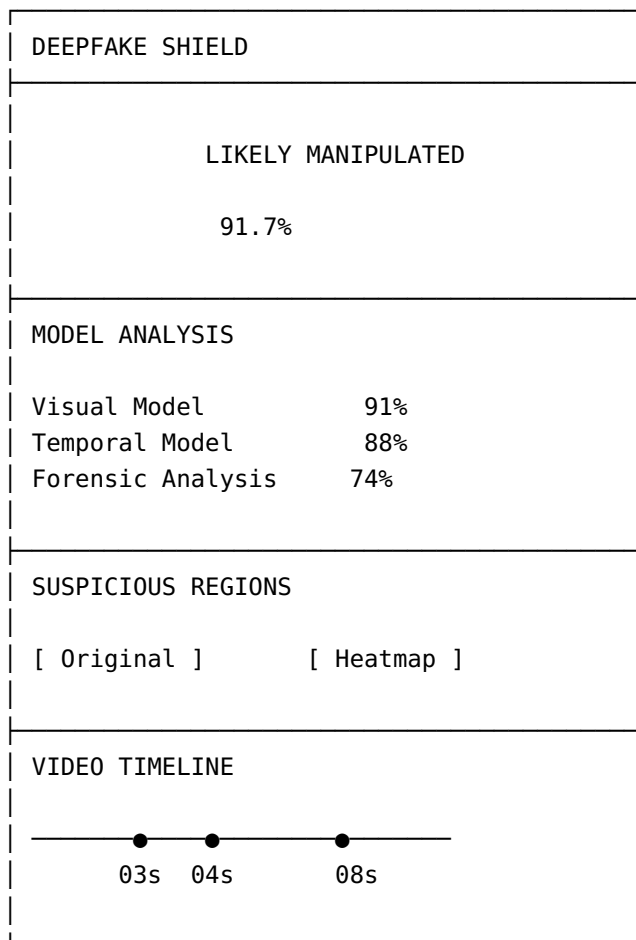
Image / Video / Audio

3. Processing

Uploading	✓
Media Analysis	✓
Face Analysis	✓
Deep Learning	🕒
Forensics	🕒
Evidence Fusion	○
Report Generation	○

24. Dashboard

The result dashboard:



25. Backend

Use:

FastAPI

Responsibilities:

- Upload handling
- File validation
- Media preprocessing
- Model inference
- Job management
- Result storage
- Report generation

26. Database

Potential database:

PostgreSQL

Tables:

```
users
analyses
media_files
model_predictions
forensic_results
suspicious_frames
evidence
reports
```

Example analysis record

Analysis ID:
DF-2026-000124

Media:
video_001.mp4

Type:
VIDEO

Result:
LIKELY_MANIPULATED

Confidence:
0.917

Model:
ViT + Temporal Transformer

Status:
COMPLETED

27. Project Structure

```
deepfake-shield/
|
├── frontend/
|   ├── app/
|   ├── components/
|   ├── pages/
|   ├── hooks/
|   └── lib/
|
├── backend/
|   ├── app/
|   |   ├── main.py
|   |   ├── api/
|   |   ├── services/
|   |   ├── models/
|   |   ├── schemas/
|   |   └── utils/
|   └── requirements.txt
|
├── ml/
|   ├── datasets/
|   ├── preprocessing/
|   └── training/
```

```
| | | inference/
| | | models/
| | | | image_model/
| | | | video_model/
| | | | audio_model/
| | |
| | | explainability/
| | | evaluation/
| |
| | forensics/
| | | metadata/
| | | compression/
| | | provenance/
| | | artifact_analysis/
| |
| | reports/
| |
| | notebooks/
| |
| | tests/
| |
| | docker/
| |
| | docs/
| |
| | README.md
```

28. Technology Stack

AI / ML

```
Python
PyTorch
OpenCV
NumPy
Pandas
Scikit-learn
TorchVision
Transformers
```

Potential models:

EfficientNet
ConvNeXt
ViT
Swin Transformer
Temporal Transformer

Explainability

Grad-CAM
Captum
Attention Visualization

Backend

FastAPI
Uvicorn
Pydantic
Celery / background workers

Database

PostgreSQL

Frontend

React / Next.js
TypeScript
Tailwind CSS

Deployment

Potential architecture:

```
Frontend
↓
Web Hosting
↓
FastAPI
↓
GPU inference server
↓
PostgreSQL
↓
Object Storage
```

For an MVP, inference can run locally.

29. API Design

Upload

```
POST /api/v1/analyze
```

Input:

```
multipart/form-data
file=<media>
```

Response:

```
{
  "analysis_id": "DF-2026-000124",
  "status": "processing"
}
```

Status

```
GET /api/v1/analyze/{analysis_id}
```

Response:

```
{  
  "status": "completed",  
  "verdict": "likely_manipulated",  
  "confidence": 0.917  
}
```

Evidence

```
GET /api/v1/analyze/{analysis_id}/evidence
```

Report

```
GET /api/v1/analyze/{analysis_id}/report
```

30. Security

Because users upload potentially sensitive media, security is important.

Implement:

- File size limits
- MIME validation
- Extension validation
- Malware scanning
- Temporary file isolation
- Secure storage
- Automatic deletion policies
- Rate limiting
- Authentication
- Authorization

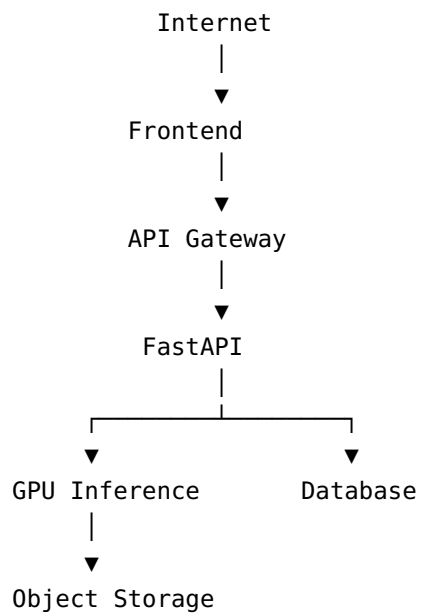
Never execute uploaded files.

31. Deployment

Development

Frontend → localhost
Backend → localhost
ML → local GPU
Database → PostgreSQL

Production



32. Development Phases

Phase 1 — Research

Study:

- Deepfakes
- GAN-generated media
- Face swapping
- Face reenactment
- Synthetic speech

- Detection methods
 - Dataset biases
-

Phase 2 — Dataset

Collect:

- Real samples
- Manipulated samples
- Multiple datasets

Perform:

- Cleaning
 - Deduplication
 - Splitting
 - Augmentation
-

Phase 3 — Baseline

Build:

```
Face Detection
  ↓
EfficientNet
  ↓
Real / Fake
```

Phase 4 — Transformer

Implement:

```
ViT
```

Compare against the baseline.

Phase 5 — Video

Implement:

```
Frame Sampling
↓
Face Tracking
↓
ViT
↓
Temporal Transformer
```

Phase 6 — Explainability

Add:

```
Grad-CAM
Attention Maps
Frame Timeline
```

Phase 7 — Forensics

Add:

```
Metadata
Compression
Artifact Detection
Provenance
```

Phase 8 — Backend

Build:

```
FastAPI
PostgreSQL
Inference Service
```

Phase 9 — Frontend

Build:

Upload
Processing
Results
Heatmap
Timeline
Forensic Report

Phase 10 — Deployment

Containerize:

Frontend
Backend
ML Service
Database

33. Team Structure

For a 3-person team:

Member 1 — Deep Learning

Responsible for:

- Dataset
 - Preprocessing
 - CNN
 - ViT
 - Video model
 - Training
 - Evaluation
-

Member 2 — Backend / ML Infrastructure

Responsible for:

- FastAPI
 - Inference pipeline
 - Database
 - File processing
 - Model serving
-

Member 3 — Frontend / UX

Responsible for:

- Dashboard
 - Upload UI
 - Heatmaps
 - Video timeline
 - Reports
 - Visualization
-

34. Research Component

The project should have a clear research question.

Main research question

How well can multimodal and temporal deep-learning models generalize to previously unseen deepfake generation techniques?

Experiment

Model A

EfficientNet

Model B

ConvNeXt

Model C

ViT

Model D

ViT + Temporal Transformer

Model E

Multimodal Fusion

Compare:

	F1	AUC	Recall
EfficientNet			
ConvNeXt			
ViT			
ViT + Temporal			
Multimodal			

35. Future Scope

Potential extensions:

- ◆ **Real-time detection**

Analyze webcam/video streams.

- ◆ **Browser extension**

Analyze suspicious media encountered online.

- ◆ **Social-media integration**

Analyze downloaded content.

- ◆ **Mobile application**

Run lightweight detection on-device.

- ◆ **Federated learning**

Train across institutions without centralizing raw media.

- ◆ **Adversarial robustness**

Evaluate performance against deliberate attempts to evade detection.

- ◆ **Model watermark detection**

Identify known AI-generation signatures where available.

- ◆ **Provenance verification**

Expand cryptographic provenance support.

- ◆ **Continuous learning**

Update models as new generation techniques appear.

36. Limitations

DeepFake Shield should explicitly acknowledge:

1. No detector can guarantee perfect detection.
2. New generation methods may evade existing models.
3. Compression can remove forensic artifacts.
4. Re-encoding may affect model performance.
5. Dataset bias can produce misleading results.
6. Metadata can be removed or manipulated.
7. Missing provenance does not prove manipulation.
8. Model confidence is not equivalent to legal certainty.

Therefore the system should present its output as:

AI-assisted forensic assessment

rather than definitive proof.

37. Final Demonstration

The final demonstration should be designed like a real forensic investigation.

Demo 1 — Authentic Image

Upload:

real_photo.jpg

System:

LIKELY AUTHENTIC

Confidence: 94%

No significant manipulation
signals detected.

Demo 2 — Manipulated Image

Upload:

face_swap.jpg

System:

LIKELY MANIPULATED

Confidence: 96%

Suspicious region:
Face boundary

Evidence:

- Visual anomaly
- Texture inconsistency
- Blending artifact

Display:

Original → Heatmap

Demo 3 — Deepfake Video

Upload:

deepfake.mp4

System:

LIKELY MANIPULATED

Confidence: 92%

Suspicious timestamps:

00:03

00:04

00:05

00:09

00:10

Clicking the timeline opens the corresponding frames.

Demo 4 — Audio

Upload:

synthetic_voice.wav

System:

LIKELY SYNTHETIC

Confidence: 87%

Spectral anomalies detected.

38. Expected Output

The final system should produce something like:

DEEPFAKE SHIELD

⚠

 LIKELY MANIPULATED

Model Confidence: 91.7%

EVIDENCE

Visual anomalies	HIGH
Temporal anomalies	HIGH
Face manipulation	HIGH
Compression anomalies	MEDIUM
Metadata signal	LOW
Provenance	UNKNOWN

SUSPICIOUS REGIONS

[HEATMAP]

SUSPICIOUS VIDEO FRAMES

00:03 ● — 00:05 ● — 00:09 ●

[VIEW FULL REPORT]

39. Success Criteria

ML

Target goals for the experimental system:

ROC-AUC	≥ 0.90
F1 Score	≥ 0.85
Recall	≥ 0.85

These are **development targets, not guaranteed outcomes**.

More important than hitting a single number is demonstrating robust performance across datasets and manipulation types.

System

- ✓ Image analysis
- ✓ Video analysis
- ✓ Audio analysis
- ✓ Explainability
- ✓ Forensic signals
- ✓ Provenance inspection
- ✓ Evidence fusion
- ✓ Web dashboard
- ✓ API
- ✓ Report generation

Demonstration

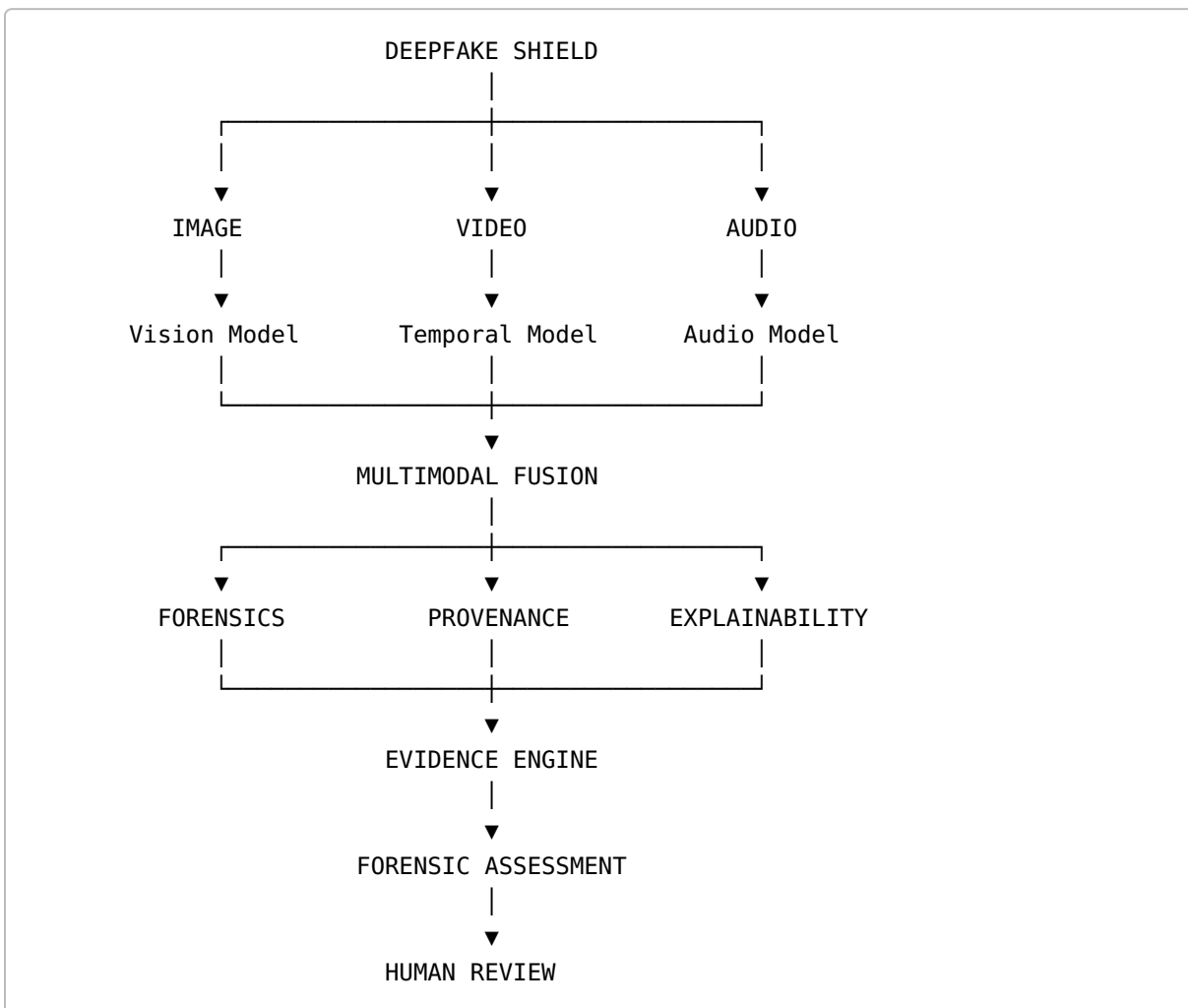
The evaluator should be able to:

```
Upload media
  ↓
Wait for analysis
  ↓
See verdict
  ↓
Inspect evidence
  ↓
View heatmap
  ↓
View suspicious frames
  ↓
Read forensic report
```

40. Final Vision

DeepFake Shield should ultimately become more than a deepfake classifier.

The long-term vision is:



The core philosophy:

AI should not simply tell you that media is fake. It should show you the evidence behind its assessment.

DeepFake Shield therefore combines:

Deep Learning + Computer Vision + Transformers + Multimodal AI + Explainable AI + Digital Forensics + Web Development

to create an end-to-end platform for **AI-assisted digital media authenticity analysis**.

 Project Tagline

DeepFake Shield

Detect. Localize. Explain. Investigate.
